Simulation of Roulette Method

Simulation Results and Performance Metrics

For the course: “Electrical Measurements and Instrumentation (ECE_621)”

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This project was developed using Python, where the simulation algorithm and the generation of graphs were created. The resulting data tables were stored using Microsoft Excel. Both the source code, the generated figures and data files alongside this report and can be found in the project repository on my GitHub at the following link: https://github.com/Georageo .

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Introduction

Queueing systems are an important area of study in the analysis of networks, computer systems, and, generally, in systems where multiple requests compete for a limited number of service resources. Understanding the behaviour of such systems is essential for evaluating their performance, estimating the probability of losses, and determining the appropriate number of resources required to provide services that satisfy our goal.

In this project, the operation of such a system is studied through simulation using the roulette method, which is based on the theory of Markov chains. This method allows the simulation of the system's states to progress by using random numbers that follow a uniform distribution. In this way, it is possible to study the behaviour of the system and estimate important performance measures, such as the probability of call loss.

To generate the random numbers, the multiplicative congruential method is used, which is a technique for producing pseudo-random numbers in computational simulations. These numbers are then used in the roulette method to simulate the transitions between the states of the system.

The System

The system used in this project is a service system of type M/M/3, according to Kendall's notation. This type of system consists of three servers that provide service to incoming requests (calls). The system operates without a waiting queue, which means that if all servers are busy when a new call arrives, the call cannot be served and is immediately rejected (lost).

The state of the system is defined by the variable **n**, which represents the number of active calls being serviced at the moment. The possible states range from 0 to 3, corresponding to the number of servers that are occupied with calls. When a call arrives and a server is free (n < 3), the system transitions from state n to state n + 1. When a service is completed, the system transitions from state n to n − 1. However, if a call arrives when the system is already in state n = 3, the call is blocked and lost, while the system remains in the same state.

- The offered traffic load in the system is 2.2 Erlang (a=2.2), which represents the average traffic caused by the incoming calls.
- The probability of the arrival of a call / request is $P_a = \frac{a}{a+n}$.
- The probability of the departure of a call / request is $P_b = \frac{n}{a+n} = 1 - P_a$.

If Y is the pseudo-random number from the uniform contribution, then:

- A call arrives ,if $0 < Y < P_a$ and the state progresses to n+1
- A call departs, if $P_a < Y < 1$ and the state progresses to n-1
- A call is lost, if all servers are occupied (n=3) and the state does not progress.

Pseudo-random Number Generation

In order to perform the simulation, a sequence of pseudorandom numbers is generated using the multiplicative congruential method:

$$x_n = k\, x_{n-1}\ (mod M)$$

where x_n is the current number, x_{n-1} the previous number, k is a constant, and M is the modulus. The first value of the algorithm, x_0, is the seed number and determines the starting point

A key property of such generators is the period, which is the number of generated values before the algorithm begins to repeat. To ensure a satisfactory long period, specific conditions must be in place. When the modulus is chosen as $M = 2^b$, the maximum possible period of the generator is 2^{b-2} (provided that the multiplier satisfies the condition $k = \pm 3 (mod 8)$ and the seed number is an odd prime number > 1.

In our case, the goal is to generate a sequence of 32 pseudorandom numbers with the maximum possible period using the multiplicative congruential generator. So, $2^{b-2} = 32$, which gives b=7. So, the modulus is $M = 2^7 = 128$. For k, we choose 3 for convenience. The initial seed number for the first simulation will be 19 (AM:1085159 , so $2 \times 9 + 1 = 19$, odd prime number).

Results

Briefing

The results of the simulation will be presented in a table containing all the necessary parameters and calculated values for each step of the algorithm. The table will include the generated pseudorandom numbers, the corresponding system states, and the resulting events (arrival, departure, or call loss). The trunks correspond to the availability vector of the servers, for example, if all servers are occupied, the vector is (1 1 1), and if all servers are available, the vector is (0 0 0).

In addition, a graphical representation of the results will be provided. The graph will illustrate the probability of call loss as a function of the seed number, including the result obtained from the current simulation as well as those from previous simulations. This comparison allows for a clearer visualization of how the estimated loss probability changes with different seed numbers and how the simulation results approach the theoretical value, which is 24%.

In the calculation of the loss probability, the first call is not taken into account, since it is always assumed to be an arrival and is therefore considered a transitional state rather than part of the system behaviour.

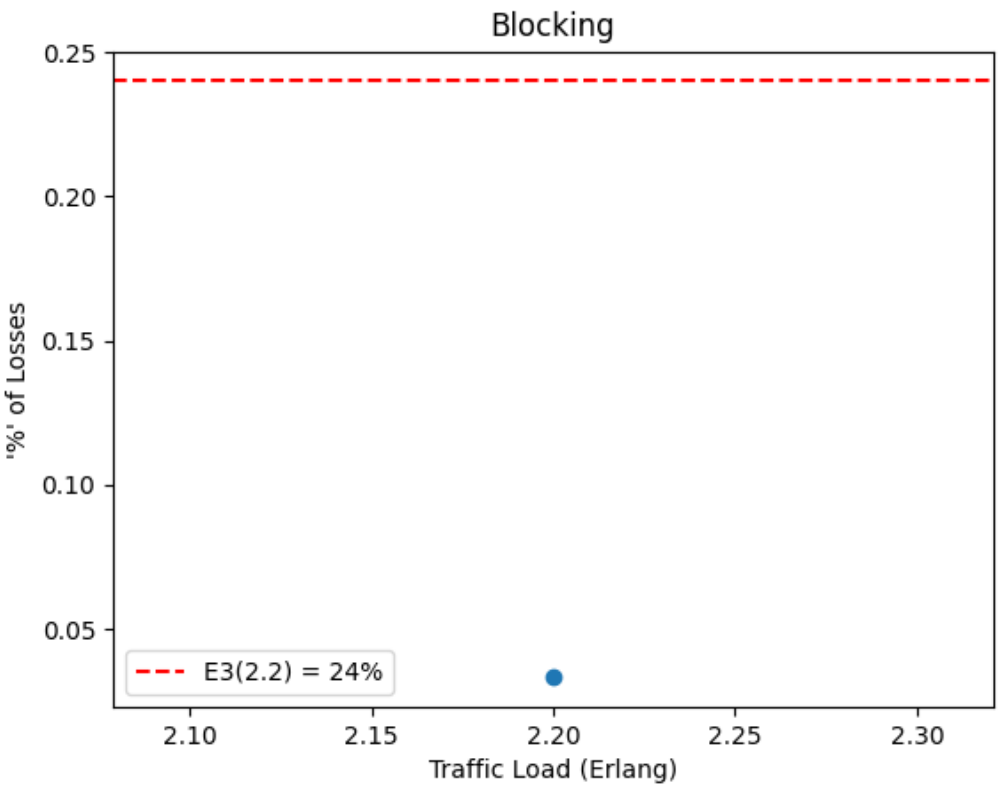
First Simulation

Data Table - seed number: 19

No	Random	Y	P(arrival)	P(departure)	State	Trunks	Service
1	57	0.43	1	0	Arrival	[1, 0, 0]	YES
2	39	0.3	0.688	0.312	Arrival	[1, 1, 0]	YES
3	117	0.89	0.524	0.476	Departure	[1, 0, 0]	YES
4	87	0.66	0.688	0.312	Arrival	[1, 1, 0]	YES
5	129	0.98	0.524	0.476	Departure	[1, 0, 0]	YES
6	123	0.93	0.688	0.312	Departure	[0, 0, 0]	YES
7	105	0.8	1	0	Arrival	[1, 0, 0]	YES
8	51	0.39	0.688	0.312	Arrival	[1, 1, 0]	YES
9	21	0.16	0.524	0.476	Arrival	[1, 1, 1]	YES
10	63	0.48	0.423	0.577	Departure	[1, 1, 0]	YES
11	57	0.43	0.524	0.476	Arrival	[1, 1, 1]	YES
12	39	0.3	0	0	Loss	[1, 1, 1]	NO
13	117	0.89	0.423	0.577	Departure	[1, 1, 0]	YES
14	87	0.66	0.524	0.476	Departure	[1, 0, 0]	YES
15	129	0.98	0.688	0.312	Departure	[0, 0, 0]	YES
16	123	0.93	1	0	Arrival	[1, 0, 0]	YES
17	105	0.8	0.688	0.312	Departure	[0, 0, 0]	YES
18	51	0.39	1	0	Arrival	[1, 0, 0]	YES
19	21	0.16	0.688	0.312	Arrival	[1, 1, 0]	YES
20	63	0.48	0.524	0.476	Arrival	[1, 1, 1]	YES
21	57	0.43	0.423	0.577	Departure	[1, 1, 0]	YES
22	39	0.3	0.524	0.476	Arrival	[1, 1, 1]	YES
23	117	0.89	0.423	0.577	Departure	[1, 1, 0]	YES
24	87	0.66	0.524	0.476	Departure	[1, 0, 0]	YES
25	129	0.98	0.688	0.312	Departure	[0, 0, 0]	YES
26	123	0.93	1	0	Arrival	[1, 0, 0]	YES
27	105	0.8	0.688	0.312	Departure	[0, 0, 0]	YES
28	51	0.39	1	0	Arrival	[1, 0, 0]	YES
29	21	0.16	0.688	0.312	Arrival	[1, 1, 0]	YES
30	63	0.48	0.524	0.476	Arrival	[1, 1, 1]	YES
31	57	0.43	0.423	0.577	Departure	[1, 1, 0]	YES
32	39	0.3	0.524	0.476	Arrival	[1, 1, 1]	YES

$$Blocking: P_{loss} = 0.0333$$

Graph



Picture 1: Graph containing the result of the first simulation

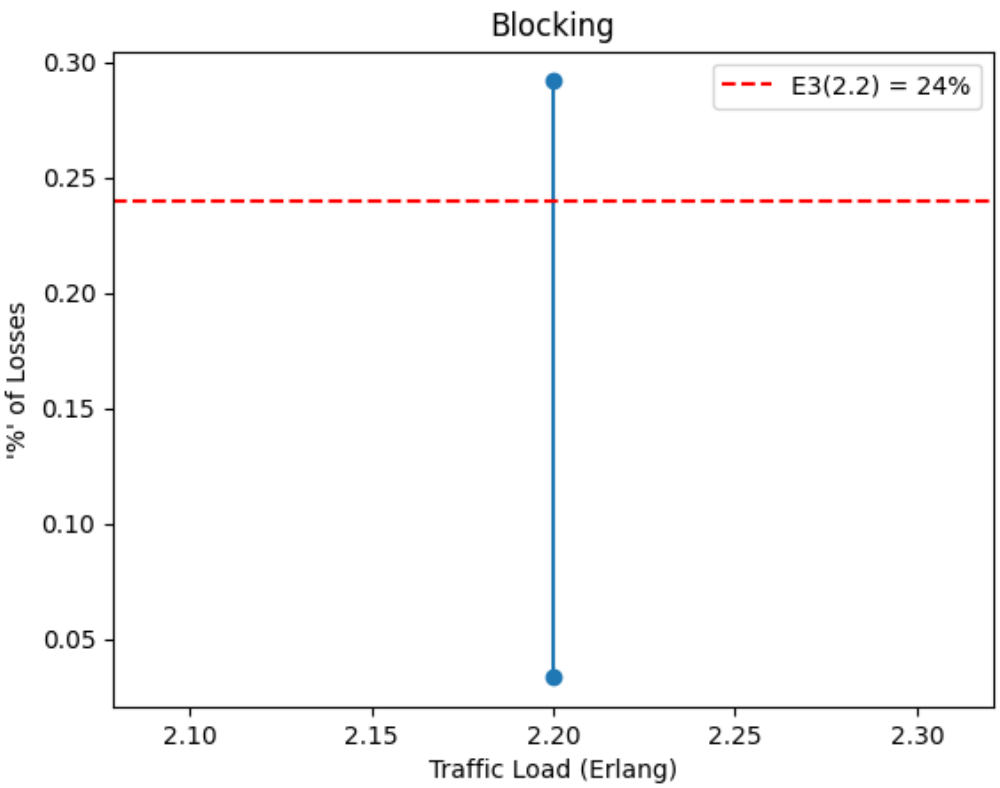
Second Simulation

Data Table – seed number: 23

No	Random	Y	P(arrival)	P(departure)	State	Trunks	Service
1	69	0.52	1	0	Arrival	[1, 0, 0]	YES
2	75	0.57	0.688	0.312	Arrival	[1, 1, 0]	YES
3	93	0.7	0.524	0.476	Departure	[1, 0, 0]	YES
4	15	0.11	0.688	0.312	Arrival	[1, 1, 0]	YES
5	45	0.34	0.524	0.476	Arrival	[1, 1, 1]	YES
6	3	0.02	0	0	Loss	[1, 1, 1]	NO
7	9	0.07	0	0	Loss	[1, 1, 1]	NO
8	27	0.2	0	0	Loss	[1, 1, 1]	NO
9	81	0.61	0.423	0.577	Departure	[1, 1, 0]	YES
10	111	0.84	0.524	0.476	Departure	[1, 0, 0]	YES
11	69	0.52	0.688	0.312	Arrival	[1, 1, 0]	YES
12	75	0.57	0.524	0.476	Departure	[1, 0, 0]	YES
13	93	0.7	0.688	0.312	Departure	[0, 0, 0]	YES
14	15	0.11	1	0	Arrival	[1, 0, 0]	YES
15	45	0.34	0.688	0.312	Arrival	[1, 1, 0]	YES
16	3	0.02	0.524	0.476	Arrival	[1, 1, 1]	YES
17	9	0.07	0	0	Loss	[1, 1, 1]	NO
18	27	0.2	0	0	Loss	[1, 1, 1]	NO
19	81	0.61	0.423	0.577	Departure	[1, 1, 0]	YES
20	111	0.84	0.524	0.476	Departure	[1, 0, 0]	YES
21	69	0.52	0.688	0.312	Arrival	[1, 1, 0]	YES
22	75	0.57	0.524	0.476	Departure	[1, 0, 0]	YES
23	93	0.7	0.688	0.312	Departure	[0, 0, 0]	YES
24	15	0.11	1	0	Arrival	[1, 0, 0]	YES
25	45	0.34	0.688	0.312	Arrival	[1, 1, 0]	YES
26	3	0.02	0.524	0.476	Arrival	[1, 1, 1]	YES
27	9	0.07	0	0	Loss	[1, 1, 1]	NO
28	27	0.2	0	0	Loss	[1, 1, 1]	NO
29	81	0.61	0.423	0.577	Departure	[1, 1, 0]	YES
30	111	0.84	0.524	0.476	Departure	[1, 0, 0]	YES
31	69	0.52	0.688	0.312	Arrival	[1, 1, 0]	YES
32	75	0.57	0.524	0.476	Departure	[1, 0, 0]	YES

$$Blocking: P_{loss} = 0.2917$$

Graph



Picture 2: Graph containing the results of the simulations 1 and 2.

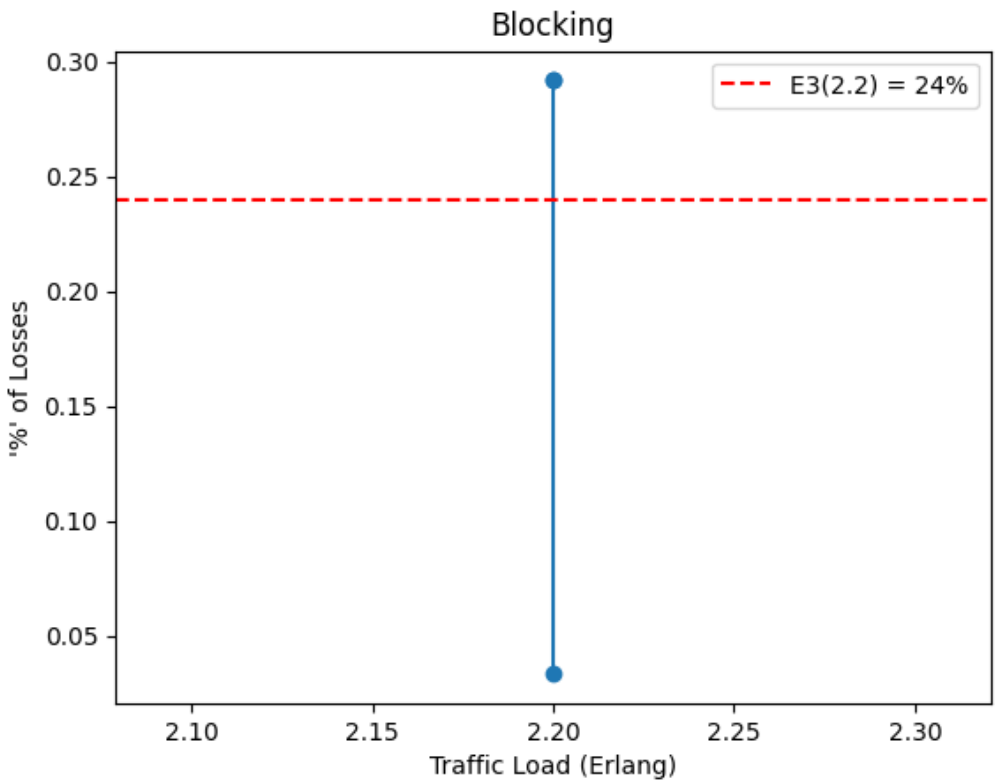
Third Simulation

Data Table – seed number: 31

No	Random	Y	P(arrival)	P(departure)	State	Trunks	Service
1	93	0.7	1	0	Arrival	[1, 0, 0]	YES
2	15	0.11	0.688	0.312	Arrival	[1, 1, 0]	YES
3	45	0.34	0.524	0.476	Arrival	[1, 1, 1]	YES
4	3	0.02	0	0	Loss	[1, 1, 1]	NO
5	9	0.07	0	0	Loss	[1, 1, 1]	NO
6	27	0.2	0	0	Loss	[1, 1, 1]	NO
7	81	0.61	0.423	0.577	Departure	[1, 1, 0]	YES
8	111	0.84	0.524	0.476	Departure	[1, 0, 0]	YES
9	69	0.52	0.688	0.312	Arrival	[1, 1, 0]	YES
10	75	0.57	0.524	0.476	Departure	[1, 0, 0]	YES
11	93	0.7	0.688	0.312	Departure	[0, 0, 0]	YES
12	15	0.11	1	0	Arrival	[1, 0, 0]	YES
13	45	0.34	0.688	0.312	Arrival	[1, 1, 0]	YES
14	3	0.02	0.524	0.476	Arrival	[1, 1, 1]	YES
15	9	0.07	0	0	Loss	[1, 1, 1]	NO
16	27	0.2	0	0	Loss	[1, 1, 1]	NO
17	81	0.61	0.423	0.577	Departure	[1, 1, 0]	YES
18	111	0.84	0.524	0.476	Departure	[1, 0, 0]	YES
19	69	0.52	0.688	0.312	Arrival	[1, 1, 0]	YES
20	75	0.57	0.524	0.476	Departure	[1, 0, 0]	YES
21	93	0.7	0.688	0.312	Departure	[0, 0, 0]	YES
22	15	0.11	1	0	Arrival	[1, 0, 0]	YES
23	45	0.34	0.688	0.312	Arrival	[1, 1, 0]	YES
24	3	0.02	0.524	0.476	Arrival	[1, 1, 1]	YES
25	9	0.07	0	0	Loss	[1, 1, 1]	NO
26	27	0.2	0	0	Loss	[1, 1, 1]	NO
27	81	0.61	0.423	0.577	Departure	[1, 1, 0]	YES
28	111	0.84	0.524	0.476	Departure	[1, 0, 0]	YES
29	69	0.52	0.688	0.312	Arrival	[1, 1, 0]	YES
30	75	0.57	0.524	0.476	Departure	[1, 0, 0]	YES
31	93	0.7	0.688	0.312	Departure	[0, 0, 0]	YES
32	15	0.11	1	0	Arrival	[1, 0, 0]	YES

$$Blocking: P_{loss} = 0.2917$$

Graph



Picture 3: Graph containing the results of simulations 1-3.

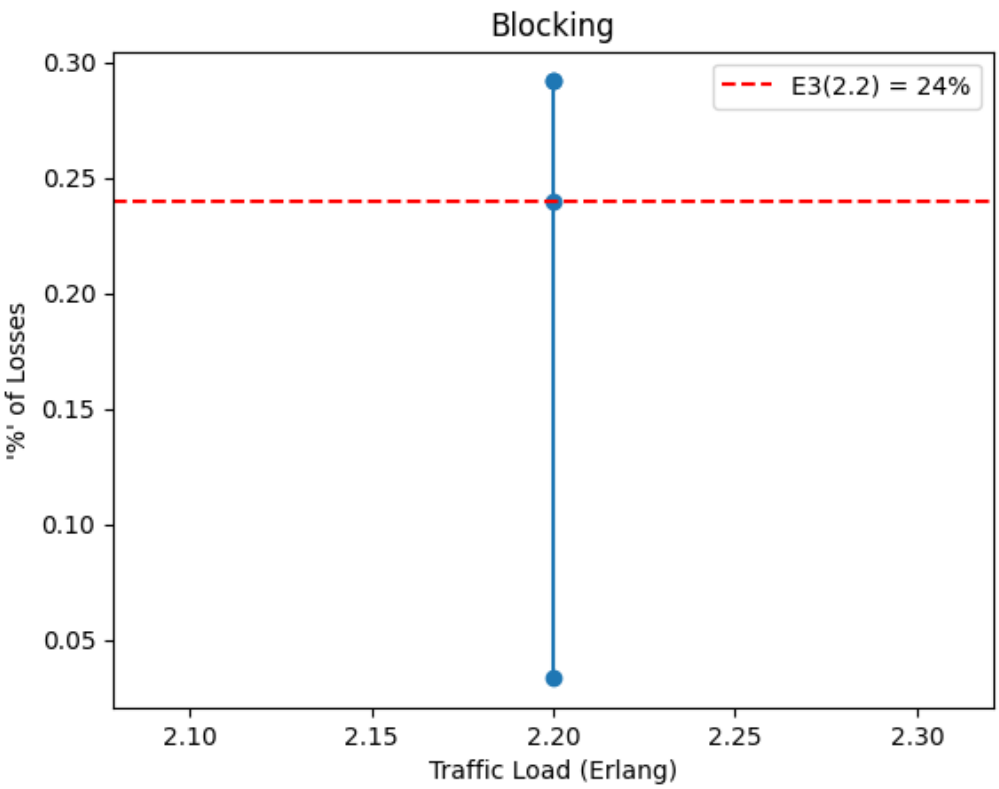
Fourth Simulation

Data Table – seed number: 37

No	Random	Y	P(arrival)	P(departure)	State	Trunks	Service
1	111	0.84	1	0	Arrival	[1, 0, 0]	YES
2	69	0.52	0.688	0.312	Arrival	[1, 1, 0]	YES
3	75	0.57	0.524	0.476	Departure	[1, 0, 0]	YES
4	93	0.7	0.688	0.312	Departure	[0, 0, 0]	YES
5	15	0.11	1	0	Arrival	[1, 0, 0]	YES
6	45	0.34	0.688	0.312	Arrival	[1, 1, 0]	YES
7	3	0.02	0.524	0.476	Arrival	[1, 1, 1]	YES
8	9	0.07	0	0	Loss	[1, 1, 1]	NO
9	27	0.2	0	0	Loss	[1, 1, 1]	NO
10	81	0.61	0.423	0.577	Departure	[1, 1, 0]	YES
11	111	0.84	0.524	0.476	Departure	[1, 0, 0]	YES
12	69	0.52	0.688	0.312	Arrival	[1, 1, 0]	YES
13	75	0.57	0.524	0.476	Departure	[1, 0, 0]	YES
14	93	0.7	0.688	0.312	Departure	[0, 0, 0]	YES
15	15	0.11	1	0	Arrival	[1, 0, 0]	YES
16	45	0.34	0.688	0.312	Arrival	[1, 1, 0]	YES
17	3	0.02	0.524	0.476	Arrival	[1, 1, 1]	YES
18	9	0.07	0	0	Loss	[1, 1, 1]	NO
19	27	0.2	0	0	Loss	[1, 1, 1]	NO
20	81	0.61	0.423	0.577	Departure	[1, 1, 0]	YES
21	111	0.84	0.524	0.476	Departure	[1, 0, 0]	YES
22	69	0.52	0.688	0.312	Arrival	[1, 1, 0]	YES
23	75	0.57	0.524	0.476	Departure	[1, 0, 0]	YES
24	93	0.7	0.688	0.312	Departure	[0, 0, 0]	YES
25	15	0.11	1	0	Arrival	[1, 0, 0]	YES
26	45	0.34	0.688	0.312	Arrival	[1, 1, 0]	YES
27	3	0.02	0.524	0.476	Arrival	[1, 1, 1]	YES
28	9	0.07	0	0	Loss	[1, 1, 1]	NO
29	27	0.2	0	0	Loss	[1, 1, 1]	NO
30	81	0.61	0.423	0.577	Departure	[1, 1, 0]	YES
31	111	0.84	0.524	0.476	Departure	[1, 0, 0]	YES
32	69	0.52	0.688	0.312	Arrival	[1, 1, 0]	YES

$$Blocking: P_{loss} = 0.24$$

Graph



Picture 4: Graph containing the results of simulations 1-4.

Mid – Simulation Analysis

In the last four simulations, the average value of the blocking probability for the offered traffic load of 2.2 (erl) was:

$$\frac{\sum_{i=1}^{4} P_{i\,losses}}{4(Number\ of\ simulations)} = \frac{(0.0333 + 2 \cdot 0.2917 + 0.24)}{4} = 0.2142$$

This average value is close to the theoretical value predicted by the Erlang B formula, suggesting that with a larger number of simulations the estimated result is to converge even more to the theoretical value.

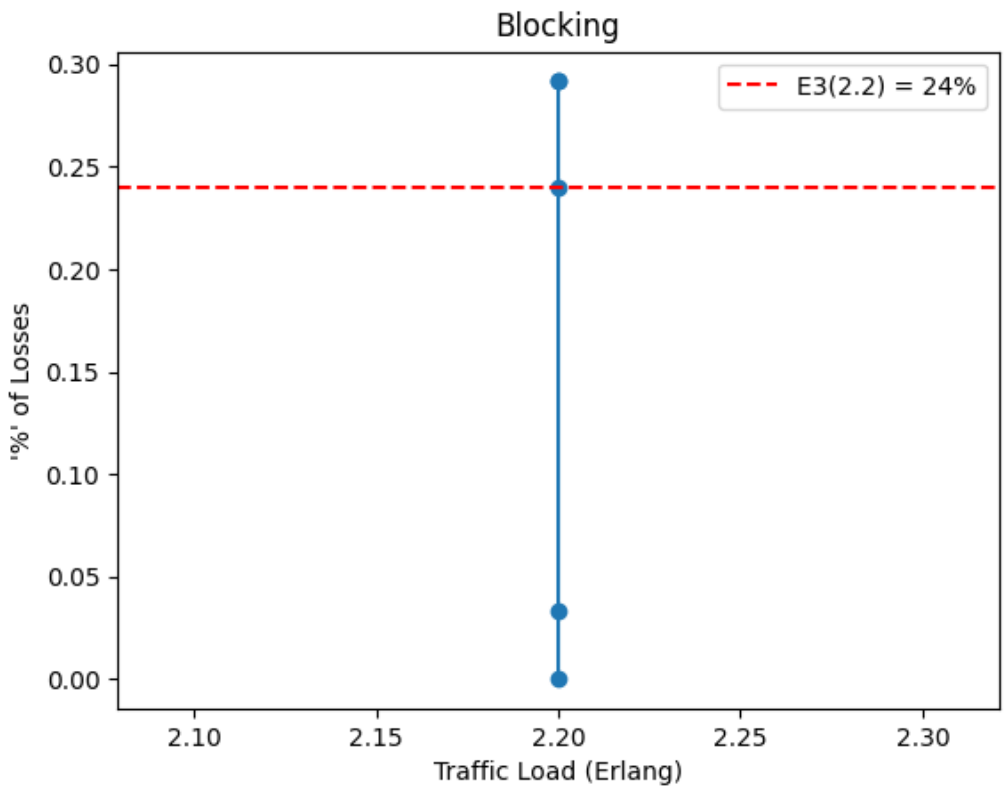
Fifth Simulation

Data Table – seed number: 41

No	Random	Y	P(arrival)	P(departure)	State	Trunks	Service
1	123	0.93	1	0	Arrival	[1, 0, 0]	YES
2	105	0.8	0.688	0.312	Departure	[0, 0, 0]	YES
3	51	0.39	1	0	Arrival	[1, 0, 0]	YES
4	21	0.16	0.688	0.312	Arrival	[1, 1, 0]	YES
5	63	0.48	0.524	0.476	Arrival	[1, 1, 1]	YES
6	57	0.43	0.423	0.577	Departure	[1, 1, 0]	YES
7	39	0.3	0.524	0.476	Arrival	[1, 1, 1]	YES
8	117	0.89	0.423	0.577	Departure	[1, 1, 0]	YES
9	87	0.66	0.524	0.476	Departure	[1, 0, 0]	YES
10	129	0.98	0.688	0.312	Departure	[0, 0, 0]	YES
11	123	0.93	1	0	Arrival	[1, 0, 0]	YES
12	105	0.8	0.688	0.312	Departure	[0, 0, 0]	YES
13	51	0.39	1	0	Arrival	[1, 0, 0]	YES
14	21	0.16	0.688	0.312	Arrival	[1, 1, 0]	YES
15	63	0.48	0.524	0.476	Arrival	[1, 1, 1]	YES
16	57	0.43	0.423	0.577	Departure	[1, 1, 0]	YES
17	39	0.3	0.524	0.476	Arrival	[1, 1, 1]	YES
18	117	0.89	0.423	0.577	Departure	[1, 1, 0]	YES
19	87	0.66	0.524	0.476	Departure	[1, 0, 0]	YES
20	129	0.98	0.688	0.312	Departure	[0, 0, 0]	YES
21	123	0.93	1	0	Arrival	[1, 0, 0]	YES
22	105	0.8	0.688	0.312	Departure	[0, 0, 0]	YES
23	51	0.39	1	0	Arrival	[1, 0, 0]	YES
24	21	0.16	0.688	0.312	Arrival	[1, 1, 0]	YES
25	63	0.48	0.524	0.476	Arrival	[1, 1, 1]	YES
26	57	0.43	0.423	0.577	Departure	[1, 1, 0]	YES
27	39	0.3	0.524	0.476	Arrival	[1, 1, 1]	YES
28	117	0.89	0.423	0.577	Departure	[1, 1, 0]	YES
29	87	0.66	0.524	0.476	Departure	[1, 0, 0]	YES
30	129	0.98	0.688	0.312	Departure	[0, 0, 0]	YES
31	123	0.93	1	0	Arrival	[1, 0, 0]	YES
32	105	0.8	0.688	0.312	Departure	[0, 0, 0]	YES

$$Blocking\text{:}\ P_{losses} = 0$$

Graph



Picture 5: Graph containing the results of simulations 1-5.

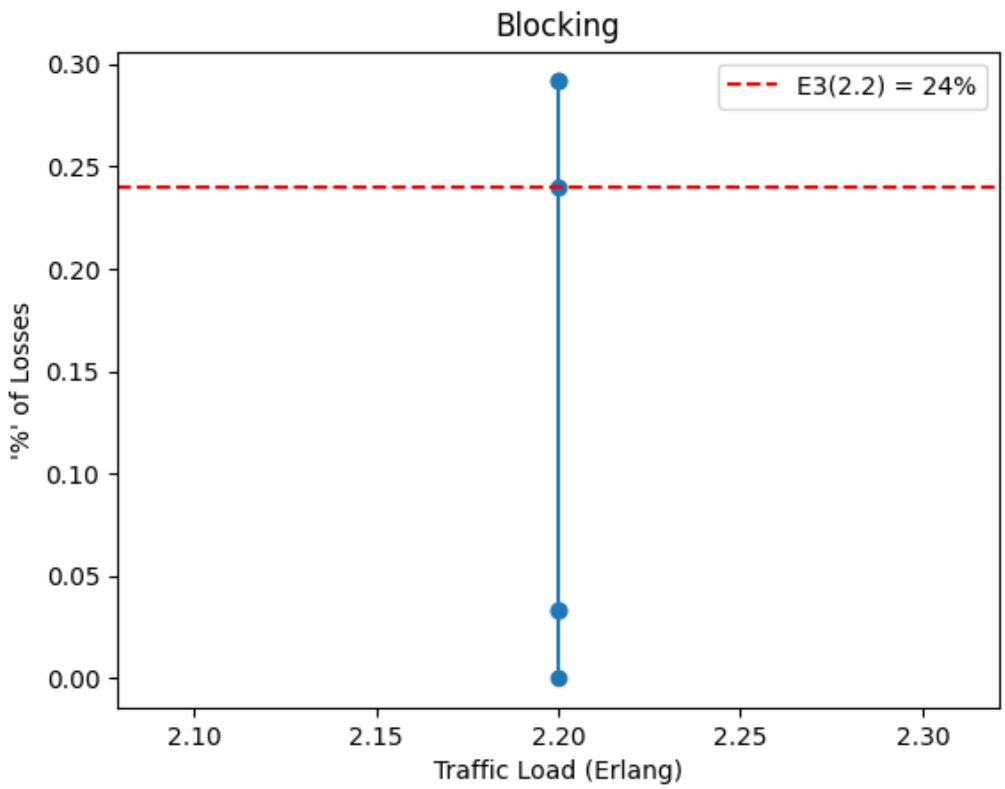
Sixth Simulation

Data Table – seed number: 43

No	Random	Y	P(arrival)	P(departure)	State	Trunks	Service
1	129	0.98	1	0	Arrival	[1, 0, 0]	YES
2	123	0.93	0.688	0.312	Departure	[0, 0, 0]	YES
3	105	0.8	1	0	Arrival	[1, 0, 0]	YES
4	51	0.39	0.688	0.312	Arrival	[1, 1, 0]	YES
5	21	0.16	0.524	0.476	Arrival	[1, 1, 1]	YES
6	63	0.48	0.423	0.577	Departure	[1, 1, 0]	YES
7	57	0.43	0.524	0.476	Arrival	[1, 1, 1]	YES
8	39	0.3	0	0	Loss	[1, 1, 1]	NO
9	117	0.89	0.423	0.577	Departure	[1, 1, 0]	YES
10	87	0.66	0.524	0.476	Departure	[1, 0, 0]	YES
11	129	0.98	0.688	0.312	Departure	[0, 0, 0]	YES
12	123	0.93	1	0	Arrival	[1, 0, 0]	YES
13	105	0.8	0.688	0.312	Departure	[0, 0, 0]	YES
14	51	0.39	1	0	Arrival	[1, 0, 0]	YES
15	21	0.16	0.688	0.312	Arrival	[1, 1, 0]	YES
16	63	0.48	0.524	0.476	Arrival	[1, 1, 1]	YES
17	57	0.43	0.423	0.577	Departure	[1, 1, 0]	YES
18	39	0.3	0.524	0.476	Arrival	[1, 1, 1]	YES
19	117	0.89	0.423	0.577	Departure	[1, 1, 0]	YES
20	87	0.66	0.524	0.476	Departure	[1, 0, 0]	YES
21	129	0.98	0.688	0.312	Departure	[0, 0, 0]	YES
22	123	0.93	1	0	Arrival	[1, 0, 0]	YES
23	105	0.8	0.688	0.312	Departure	[0, 0, 0]	YES
24	51	0.39	1	0	Arrival	[1, 0, 0]	YES
25	21	0.16	0.688	0.312	Arrival	[1, 1, 0]	YES
26	63	0.48	0.524	0.476	Arrival	[1, 1, 1]	YES
27	57	0.43	0.423	0.577	Departure	[1, 1, 0]	YES
28	39	0.3	0.524	0.476	Arrival	[1, 1, 1]	YES
29	117	0.89	0.423	0.577	Departure	[1, 1, 0]	YES
30	87	0.66	0.524	0.476	Departure	[1, 0, 0]	YES
31	129	0.98	0.688	0.312	Departure	[0, 0, 0]	YES
32	123	0.93	1	0	Arrival	[1, 0, 0]	YES

$$Blocking: P_{loss} = 0.0333$$

Graph



Picture 6: Graph containing the results of simulations 1-6.

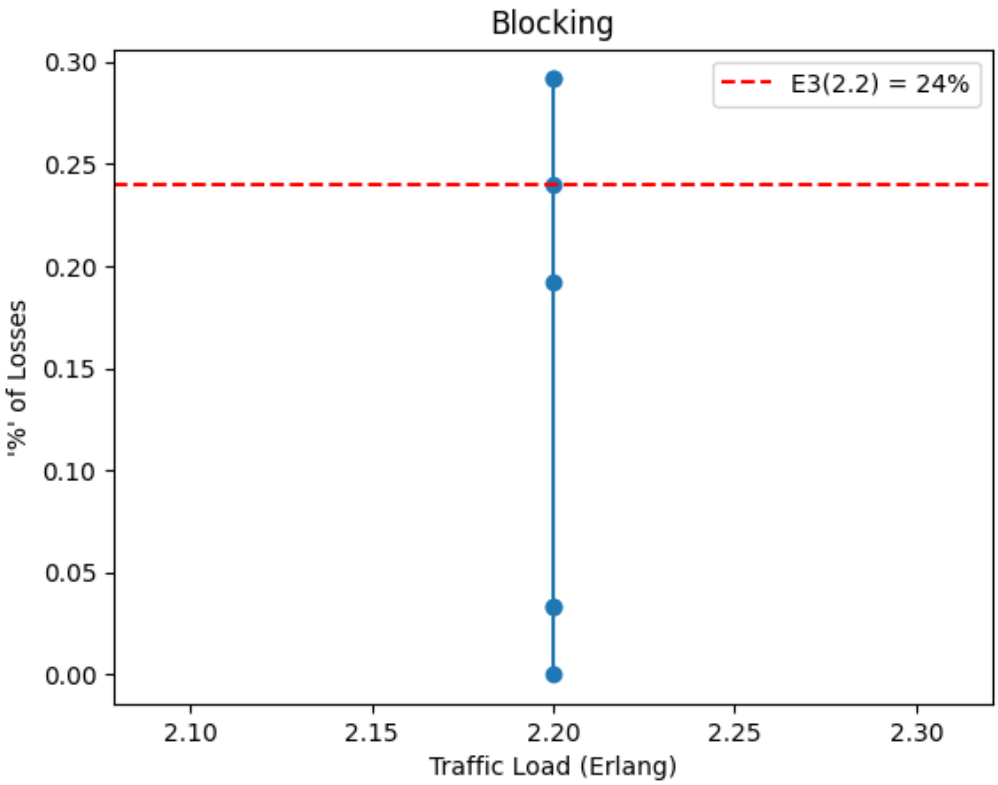
Seventh Simulation

Data Table – seed number: 59

No	Random	Y	P(arrival)	P(departure)	State	Trunks	Service
1	45	0.34	1	0	Arrival	[1, 0, 0]	YES
2	3	0.02	0.688	0.312	Arrival	[1, 1, 0]	YES
3	9	0.07	0.524	0.476	Arrival	[1, 1, 1]	YES
4	27	0.2	0	0	Loss	[1, 1, 1]	NO
5	81	0.61	0.423	0.577	Departure	[1, 1, 0]	YES
6	111	0.84	0.524	0.476	Departure	[1, 0, 0]	YES
7	69	0.52	0.688	0.312	Arrival	[1, 1, 0]	YES
8	75	0.57	0.524	0.476	Departure	[1, 0, 0]	YES
9	93	0.7	0.688	0.312	Departure	[0, 0, 0]	YES
10	15	0.11	1	0	Arrival	[1, 0, 0]	YES
11	45	0.34	0.688	0.312	Arrival	[1, 1, 0]	YES
12	3	0.02	0.524	0.476	Arrival	[1, 1, 1]	YES
13	9	0.07	0	0	Loss	[1, 1, 1]	NO
14	27	0.2	0	0	Loss	[1, 1, 1]	NO
15	81	0.61	0.423	0.577	Departure	[1, 1, 0]	YES
16	111	0.84	0.524	0.476	Departure	[1, 0, 0]	YES
17	69	0.52	0.688	0.312	Arrival	[1, 1, 0]	YES
18	75	0.57	0.524	0.476	Departure	[1, 0, 0]	YES
19	93	0.7	0.688	0.312	Departure	[0, 0, 0]	YES
20	15	0.11	1	0	Arrival	[1, 0, 0]	YES
21	45	0.34	0.688	0.312	Arrival	[1, 1, 0]	YES
22	3	0.02	0.524	0.476	Arrival	[1, 1, 1]	YES
23	9	0.07	0	0	Loss	[1, 1, 1]	NO
24	27	0.2	0	0	Loss	[1, 1, 1]	NO
25	81	0.61	0.423	0.577	Departure	[1, 1, 0]	YES
26	111	0.84	0.524	0.476	Departure	[1, 0, 0]	YES
27	69	0.52	0.688	0.312	Arrival	[1, 1, 0]	YES
28	75	0.57	0.524	0.476	Departure	[1, 0, 0]	YES
29	93	0.7	0.688	0.312	Departure	[0, 0, 0]	YES
30	15	0.11	1	0	Arrival	[1, 0, 0]	YES
31	45	0.34	0.688	0.312	Arrival	[1, 1, 0]	YES
32	3	0.02	0.524	0.476	Arrival	[1, 1, 1]	YES

$$Blocking: P_{loss} = 0.1923$$

Graph



Picture 7: Graph containing the results of simulations 1-7.

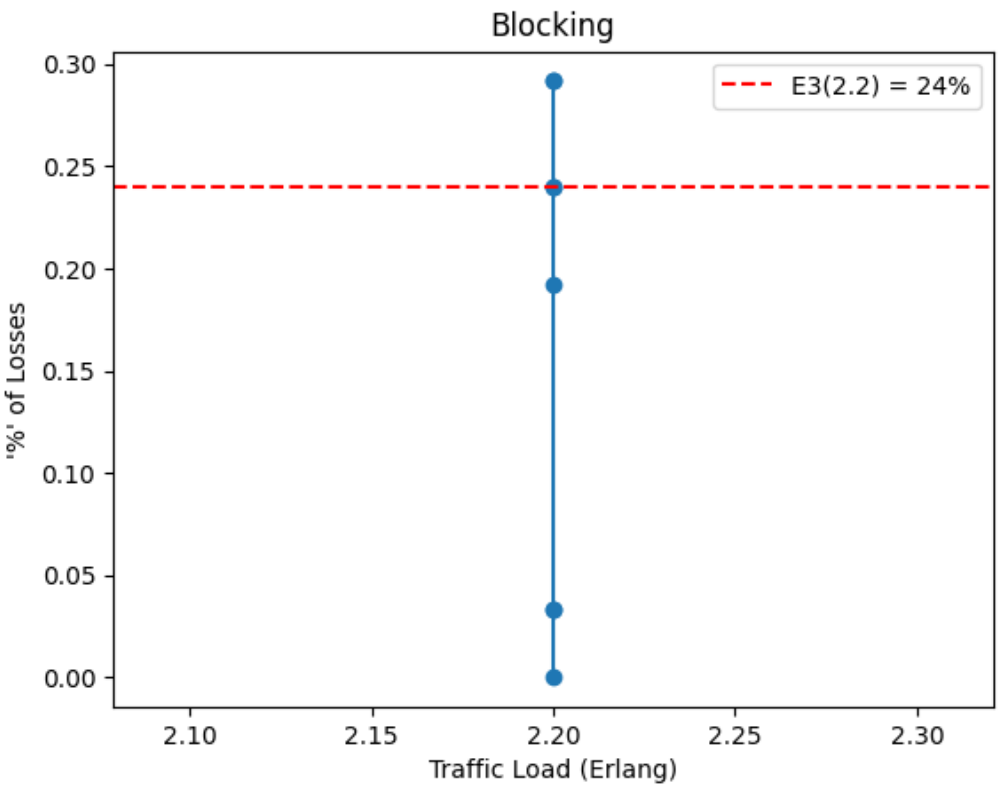
Eighth Simulation

Data Table – seed number: 97

No	Random	Y	P(arrival)	P(departure)	State	Trunks	Service
1	27	0.2	1	0	Arrival	[1, 0, 0]	YES
2	81	0.61	0.688	0.312	Arrival	[1, 1, 0]	YES
3	111	0.84	0.524	0.476	Departure	[1, 0, 0]	YES
4	69	0.52	0.688	0.312	Arrival	[1, 1, 0]	YES
5	75	0.57	0.524	0.476	Departure	[1, 0, 0]	YES
6	93	0.7	0.688	0.312	Departure	[0, 0, 0]	YES
7	15	0.11	1	0	Arrival	[1, 0, 0]	YES
8	45	0.34	0.688	0.312	Arrival	[1, 1, 0]	YES
9	3	0.02	0.524	0.476	Arrival	[1, 1, 1]	YES
10	9	0.07	0	0	Loss	[1, 1, 1]	NO
11	27	0.2	0	0	Loss	[1, 1, 1]	NO
12	81	0.61	0.423	0.577	Departure	[1, 1, 0]	YES
13	111	0.84	0.524	0.476	Departure	[1, 0, 0]	YES
14	69	0.52	0.688	0.312	Arrival	[1, 1, 0]	YES
15	75	0.57	0.524	0.476	Departure	[1, 0, 0]	YES
16	93	0.7	0.688	0.312	Departure	[0, 0, 0]	YES
17	15	0.11	1	0	Arrival	[1, 0, 0]	YES
18	45	0.34	0.688	0.312	Arrival	[1, 1, 0]	YES
19	3	0.02	0.524	0.476	Arrival	[1, 1, 1]	YES
20	9	0.07	0	0	Loss	[1, 1, 1]	NO
21	27	0.2	0	0	Loss	[1, 1, 1]	NO
22	81	0.61	0.423	0.577	Departure	[1, 1, 0]	YES
23	111	0.84	0.524	0.476	Departure	[1, 0, 0]	YES
24	69	0.52	0.688	0.312	Arrival	[1, 1, 0]	YES
25	75	0.57	0.524	0.476	Departure	[1, 0, 0]	YES
26	93	0.7	0.688	0.312	Departure	[0, 0, 0]	YES
27	15	0.11	1	0	Arrival	[1, 0, 0]	YES
28	45	0.34	0.688	0.312	Arrival	[1, 1, 0]	YES
29	3	0.02	0.524	0.476	Arrival	[1, 1, 1]	YES
30	9	0.07	0	0	Loss	[1, 1, 1]	NO
31	27	0.2	0	0	Loss	[1, 1, 1]	NO
32	81	0.61	0.423	0.577	Departure	[1, 1, 0]	YES

$$Blocking: P_{loss} = 0.24$$

Graph



Picture 8: Graph containing the results of simulations 1-8.

Simulation Aftermath Analysis

During the simulations, the average value of the blocking probability for the offered traffic load of 2.2 (erl) was:

$$\frac{\sum_{i=1}^{8} P_{i\,losses}}{8(Number\ of\ simulations)} = \frac{(2 \cdot 0.0333 + 2 \cdot 0.2917 + 2 \cdot 0.24 + 0.1923)}{8} = 0.1653$$

This number is different from the theoretical value mainly because some simulations resulted in very few (first and sixth) or zero (fifth) losses, and because the number of simulation steps used was relatively small, which increases the effect of randomness on the estimated blocking probability.

Conclusions

The simulation results provide an estimation of the call loss probability for an M/M/3 system with a traffic load of 2.2 Erlang. The values obtained from the simulations vary depending on the seed number used for the number generator, since each one produces a different list of random numbers and so, a different sequence of state transitions of the system. As a result, the estimated loss probability may change between simulations and may also deviate from the theoretical value given by the Erlang B formula (24%), especially because the simulation is based on a relatively small number of repetitions (32 steps). We can clearly see that in the difference between the graph of first 4 (Picture 4) and the graph of the complete simulation (Picture 8) , in which the second (0.2917), third (0.2917), **fourth (0.24)**, seventh(0.1923) and **last (0.24)** simulation approach the value of 0.24 with the other ones deviating a lot.